Bayesian Econometrics

Week 3, Summer Term 2026

WU Vienna University of Economics and Business

Today

Model selection

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Model selection

The Bayesian Approach to Model Selection

- K models $\mathcal{M}_1, \ldots, \mathcal{M}_K$ are provided to explain observations $\mathbf{y} = (y_1, \ldots, y_N)$.

- Each model $\mathcal{M}_k$ is characterized by
  - a vector of unknown parameters $\boldsymbol{\theta}_k$;
  - the density $p(\mathbf{y}|\boldsymbol{\theta}_k, \mathcal{M}_k)$ of the conditional probability distribution of $\mathbf{y}$ given $\boldsymbol{\theta}_k$;
  - the density $p(\boldsymbol{\theta}_k|\mathcal{M}_k)$ of the prior distribution of $\boldsymbol{\theta}_k$.

- Consider each model under consideration as arising from a discrete model space $\mathcal{M}_1, \ldots, \mathcal{M}_K$.

The Bayesian Approach to Model Selection

- Assign **prior probabilities** $\mathrm{Pr}(\mathcal{M}_1), \ldots, \mathrm{Pr}(\mathcal{M}_K)$ to each model (choose, for instance, a uniform distribution over the model space)
- Given observations $\mathbf{y} = (y_1, \ldots, y_N)$, compute the **posterior probability** distribution over the model space, i.e. $\mathrm{Pr}(\mathcal{M}_1|\mathbf{y}), \ldots, \mathrm{Pr}(\mathcal{M}_K|\mathbf{y})$, using Bayes' rule:

$$\mathrm{Pr}(\mathcal{M}_k|\mathbf{y}) \propto p(\mathbf{y}|\mathcal{M}_k)\mathrm{Pr}(\mathcal{M}_k), \qquad k = 1, \ldots, K.$$

- $p(\mathbf{y}|\mathcal{M}_k)$ is the **marginal likelihood** of model $\mathcal{M}_k$:

$$p(\mathbf{y}|\mathcal{M}_k) = \int_{\Theta_k} p(\mathbf{y}, \boldsymbol{\theta}_k|\mathcal{M}_k)d\boldsymbol{\theta}_k = \int_{\Theta_k} p(\mathbf{y}|\boldsymbol{\theta}_k, \mathcal{M}_k)p(\boldsymbol{\theta}_k|\mathcal{M}_k)d\boldsymbol{\theta}_k.$$

Decision-theoretic approach to model selection

The 0/1 loss function

Consider the 0/1 loss function

$$R(\mathcal{M}_{\hat{k}}, \mathcal{M}_k) = \begin{cases} 0, & \hat{k} = k, \\ 1, & \hat{k} \neq k. \end{cases}$$

all other models are equally wrong

correct model

- Choosing the model $\mathcal{M}_{\hat{k}}$ with the highest posterior probability $\Pr(\mathcal{M}_{\hat{k}}|\mathbf{y})$ will minimize the expected risk.
- If all models have the same prior probability $\Pr(\mathcal{M}_k) = 1/K$, this strategy results in selecting the model $\mathcal{M}_{\hat{k}}$ with the highest marginal likelihood $p(\mathbf{y}|\mathcal{M}_{\hat{k}})$.

Computing marginal likelihoods

Exact Computation of the Marginal Likelihood

- Bayes' theorem revisited:

$$p(\boldsymbol{\theta}_k|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)}{\int p(\mathbf{y}|\boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)d\boldsymbol{\theta}_k} = \frac{p(\mathbf{y}|\boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)}{p(\mathbf{y}|\mathcal{M}_k)}.$$

- **The marginal likelihood is equal to the normalizing constant of the non-normalized posterior density**.

- **Candidate's formula [Besag, 1989]** The marginal likelihood may be computed explicitly, if the posterior density arises from a well-known distribution family :

$$p(\mathbf{y}|\mathcal{M}_k) = \frac{p(\mathbf{y}|\boldsymbol{\theta}_k)p(\boldsymbol{\theta}_k)}{p(\boldsymbol{\theta}_k|\mathbf{y})}. \tag{1}$$

Note: (1) holds for arbitrary values $\boldsymbol{\theta}_k$.

Bayesian model selection for regression models

- For the standard linear regression model with unknown homoscedastic variance σ^2 under the **conjugate prior**:

$$\beta|\sigma^2 \sim \mathcal{N}_d\left(\mathbf{b}_0, \sigma^2\mathbf{B}_0\right), \qquad \sigma^2 \sim \mathcal{G}^{-1}\left(c_0, C_0\right),$$

the joint posterior density of (β, σ^2) is of closed form:

$$\beta|\sigma^2, \mathbf{y} \sim \mathcal{N}_d\left(\mathbf{b}_N, \sigma^2\mathbf{B}_N\right), \qquad \sigma^2|\mathbf{y} \sim \mathcal{G}^{-1}\left(c_N, C_N\right).$$

$\Rightarrow$ The marginal likelihood is available explicitly for any proper prior with $|\mathbf{B}_0| \neq 0$, $C_0 > 0$ and $c_0 > 0$:

$$p(\mathbf{y}|\mathcal{M}) = |\mathbf{B}_0^{-1}|^{\frac{1}{2}} \frac{(C_0)^{c_0}}{\Gamma(c_0)} \times \frac{\Gamma(c_N)|\mathbf{B}_N|^{\frac{1}{2}}}{(C_N)^{c_N}} \times \frac{1}{(2\pi)^{\frac{N}{2}}}.$$

Chib's estimator of the marginal Likelihood

- For a regression model with non-conjugate prior with $\vartheta_k = (\beta, \sigma^2)$:

$$\hat{p}(\beta, \sigma^2|\mathbf{y}) \quad = \quad p(\sigma^2|\mathbf{y}, \beta)\hat{p}(\beta|\mathbf{y}).$$

- The posterior ordinate $\hat{p}(\beta|\mathbf{y})$ is estimated from the MCMC draws:

$$p(\beta|\mathbf{y}) = \int p(\beta|\sigma^2, \mathbf{y})p(\sigma^2|\mathbf{y})d\,\sigma^2 \Rightarrow \hat{p}(\beta|\mathbf{y}) = \frac{1}{M}\sum_{m=1}^{M} p(\beta|\sigma^{2,(m)}, \mathbf{y}).$$

- **Chib's estimator** [Chib, 1995]

$$\hat{p}_{CH}(\mathbf{y}|\mathcal{M}_k) = \frac{p(\mathbf{y}|\boldsymbol{\theta}_k^\star)p(\boldsymbol{\theta}_k^\star)}{\hat{p}(\boldsymbol{\theta}_k^\star|\mathbf{y}, \mathcal{M}_k)},$$

very volatile estimator (depends a lot on the exact theta plugged into the estimator)

where $\boldsymbol{\theta}_k^\star$ is a high posterior point (e.g. the posterior mean or the posterior mode).

Bayes factor

- The odds ratio of model $\mathcal{M}_1$ versus $\mathcal{M}_2$ is called the **Bayes factor**:

$$\mathrm{BF} = B(\mathbf{y}) = \frac{p(\mathbf{y}|\mathcal{M}_1)}{p(\mathbf{y}|\mathcal{M}_2)}.$$

- Interpretation of the Bayes factor (Kass and Raftery):

$B(\mathbf{y})$	Interpretation
$[1, 3[$	worth bare mention
$[3, 20[$	positive
$[20, 150[$	strong
≥ 150	very strong

- A Bayes factor < 1 supports $\mathcal{M}_2$.
- **Lindley's paradox:** As the prior is becoming more and more vague, the simpler model will be chosen with probability 1, regardless of the data $\mathbf{y}$, the sample size N, and the true model.

Bayesian Predictive Analysis

Bayesian Predictive Analysis

Use Bayesian inference to forecast future outcomes $\mathbf{y}_f$,

- from **data** $\mathbf{y} = (y_1, \ldots, y_N)$,

using an appropriate **model**, e.g.

- for $\mathrm{i.i.d.}$ data, a model based on $p(y_i \mid \boldsymbol{\theta})$;
- for cross-sectional data, a model based on $p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta})$;
- in a time series context, a model based on $p(y_t \mid y^{t-1}, \boldsymbol{\theta})$, where $y^{t-1} = (y_1, \ldots, y_{t-1})$.

Apply Bayesian principle to derive $p(\textit{latent} \mid \textit{observed})$, i.e. $p(\mathbf{y}_f \mid \mathbf{y})$.

we now use Bayes rule again but not for parameters but for future observations

The posterior predictive distribution

- The **posterior predictive distribution** of $\mathbf{y}_f$ conditional on data $\mathbf{y}$ is given as

$$p(\mathbf{y}_f \mid \mathbf{y}) = \int_\Theta p(\mathbf{y}_f, \boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta} = \int_\Theta \underbrace{p(\mathbf{y}_f \mid \boldsymbol{\theta}, \mathbf{y})}_{\text{sampling dist.}} \underbrace{p(\boldsymbol{\theta} \mid \mathbf{y})}_{\text{posterior to give the "weight" of theta}} d\boldsymbol{\theta}$$

- Bayesian forecasting yields the **entire** predictive distribution.
- It depends on the observed data, but not on unknown quantities - it is obtained from $p(y_f, \theta|\mathbf{y})$ marginalized over the parameter $\boldsymbol{\theta}$.
- The predictive distribution is the 'expectation' of $p(y_f|\boldsymbol{\theta})$ with respect to the posterior of $\boldsymbol{\theta}$.

Sampling from the posterior predictive distribution

- The posterior predictive distribution can be approximated by MC methods.
- **Algorithm:**

  For $m = 1, \dots, M$:
  - Sample $\boldsymbol{\theta}^{(m)}$ from the posterior $\boldsymbol{\theta} \sim p(\boldsymbol{\theta}|y_1, \dots, y_n)$.
  - Sample y_f from the likelihood

  $$y_f^{(m)} \sim p(y_f|\boldsymbol{\theta}^{(m)})$$

- Note that $(\boldsymbol{\theta}, y_f)^{(1)}, \dots, (\boldsymbol{\theta}, y_f)^{(M)}$ are M independent draws from the *joint posterior*

  $$p(y_f, \boldsymbol{\theta}|y_1, \dots, y_n),$$

- while $(y_1^{(1)}, \dots, y_f^{(M)})$ are M independent draws from the posterior predictive distribution

  $$p(y_f|y_1, \dots, y_n).$$

The posterior predictive distribution of a time series

- Consider **H-step ahead predictions** $\mathbf{y}_f = (y_{N+1}, \ldots, y_{N+H})$ of a time series $\mathbf{y} = (y_1, \ldots, y_N)$.
- The **posterior predictive density** $p(\mathbf{y}_f \mid \mathbf{y})$ is given by:

$$p(\mathbf{y}_f \mid \mathbf{y}) = \int p(\mathbf{y}_f \mid \boldsymbol{\theta}, \mathbf{y}) p(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta}.$$

where

$$p(\mathbf{y}_f \mid \theta, \mathbf{y}) = p(y_{N+1} \mid \mathbf{y}, \theta) p(y_{N+2} \mid y_{N+1}, \mathbf{y}, \theta) \prod_{h=3}^{H} p(y_{N+h} \mid y_{N+1}, \ldots, y_{N+h-1}, \mathbf{y}, \theta).$$

A sampling-based approach to prediction

- A lot of integrals, if **marginal *h*-step** ahead density $p(y_{N+h} \mid \mathbf{y})$ is of interest.
- **Monte Carlo approach:** if a sample $(\mathbf{y}_f^{(m)}, \boldsymbol{\theta}^{(m)})$, $m = 1, \ldots, M$ from the joint distribution $p(\mathbf{y}_f, \boldsymbol{\theta} \mid \mathbf{y})$ is available, then
    - the reduced sample $\mathbf{y}_f^{(m)}$, $m = 1, \ldots, M$, is a sample from the marginal distribution $p(\mathbf{y}_f \mid \mathbf{y})$
    - the reduced sample $y_{N+h}^{(m)}$, $m = 1, \ldots, M$, is a sample from the marginal distribution $p(y_{N+h} \mid \mathbf{y})$.
- These samples can be used to approximate the distributions $p(\mathbf{y}_f \mid \mathbf{y})$ and $p(y_{N+h} \mid \mathbf{y})$.
- Works also for multivariate time series.

A sampling based approach to time series prediction

Sample from the joint distribution $p(\mathbf{y}_f, \boldsymbol{\theta} \mid \mathbf{y}) = p(\mathbf{y}_f \mid \boldsymbol{\theta}, \mathbf{y}) p(\boldsymbol{\theta} \mid \mathbf{y})$**:**

(a) sample $\boldsymbol{\theta}^{(m)}$ from the posterior density $p(\boldsymbol{\theta} \mid \mathbf{y})$;

(b) sample $\mathbf{y}_f^{(m)}$ from the *conditional* distribution $p(\mathbf{y}_f \mid \boldsymbol{\theta}^{(m)}, \mathbf{y})$.

- Step (b): **Dynamic forecasting** to sample future paths $\mathbf{y}_f^{(m)} = (y_{N+1}^{(m)}, \dots, y_{N+H}^{(m)})$:
  - for $h = 1$, sample $y_{N+1}^{(m)}$ from $p(y_{N+1} \mid \boldsymbol{\theta}^{(m)}, \mathbf{y})$;
  - for $h = 2$, sample $y_{N+2}^{(m)}$ from $p(y_{N+2} \mid y_{N+1}^{(m)}, \mathbf{y}, \boldsymbol{\theta}^{(m)})$;
  - for $h = 3$, sample $y_{N+3}^{(m)}$ from $p(y_{N+3} \mid y_{N+2}^{(m)}, y_{N+1}^{(m)}, \mathbf{y}, \boldsymbol{\theta}^{(m)})$;
  - for $h > 3$, sample $y_{N+h}^{(m)}$ from $p(y_{N+h} \mid y_{N+h-1}^{(m)}, \dots, y_{N+1}^{(m)}, \mathbf{y}, \boldsymbol{\theta}^{(m)})$.

Model assessment

- Assessing model fit is an important issue in statistical inference.
- In the Bayesian approach model assessment can be accomplished by investigating discrepancies between
  - the empirical distribtion of the data $\mathbf{y}$ and
  - the posterior predictive distribution of samples $\mathbf{y}_f$ of the same size n.
- If the model is appropriate, data sets $\mathbf{y}_f$ from the posterior predictive distribution should resemble $\mathbf{y}$.

$\Rightarrow$ Posterior predictive model checking can be performed by sampling data sets from the posterior predictive distribution.

Posterior predictive model checking

Model assessment by sampling from the posterior predictive distribution:

- Choose a function $T(\mathbf{y})$ that captures (relevant) features of the data:
    - $\mathbf{y}$ is the observed data of size n.
    - $T(\mathbf{y})$ can be a (lower dimensional) statistic computed from the data (or the data set itself, i.e. $T(\mathbf{y}) = \mathbf{y}$).
- Determine the posterior predictive distribution of $T(\mathbf{y})$:
    - Sample $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(M)}$ from the posterior distribution.
    - For $m = 1, \ldots, M$:
        - Generate a posterior predictive data set $\mathbf{y}_f^{(m)}$ of size n from $p(\mathbf{y}_f | \boldsymbol{\theta}^{(m)})$.
        - Determine $T(\mathbf{y}_f^{(m)})$.
- Compare the statistic of the observed data $T(\mathbf{y})$ to the posterior predictive distribution $p(T(\mathbf{y}_f)|\mathbf{y})$.

Assessing model misspecification i

- Model misspecification is indicated if $T(\mathbf{y})$ is in the tails of the posterior predictive distribution $p(T(\mathbf{y}_f)|\mathbf{y})$.
- The evidence for model misspecification can be quantified by the **posterior predictive p-value** (Bayesian p-value).
- If large values of $p(T(\mathbf{y}_f)|\mathbf{y})$ indicate model misspecification, the Bayesian posterior predictive p-value is given as

$$p_B = P(T(\mathbf{y}_f) > T(\mathbf{y})|\mathbf{y})$$

- Idea: predictive data sets $\mathbf{y}_f$ should ressemble the observed data $\mathbf{y}$ with respect to features of interest.

Assessing model misspecification ii

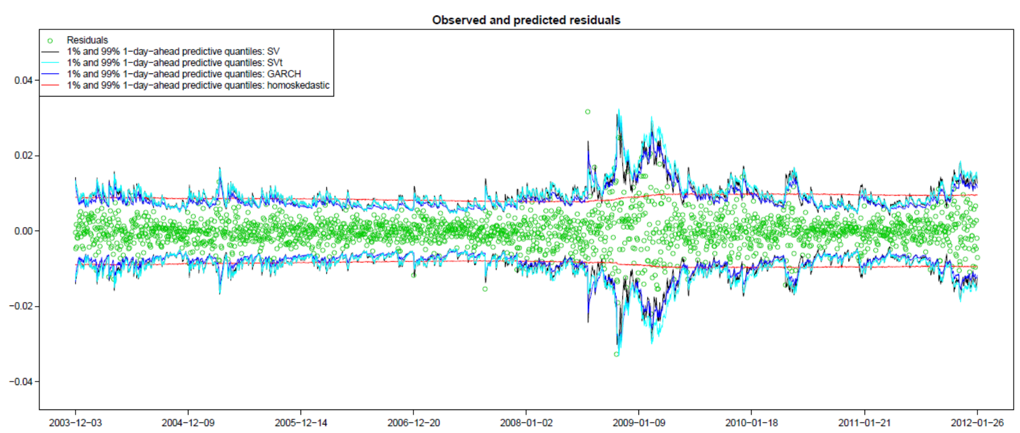
- p_B is the probability that the replicated data could be more extreme than the observed data, as measured by the test quantity T.
- Interpretation: if p_B is close to 0 or 1, e.g. outside of the interval $(0.05, 0.95)$, then the model cannot capture the aspect of the data measured by T.
- Choosing test quantities: it is good practice to apply multiple functions T.

Assessing model misspecification iii

- Computation via simulation: assume we already have M simulations $\vartheta^{(1)}, \dots, \vartheta^{(M)}$ from $p(\vartheta \mid y)$, then:
  1. sample fake data $y^{\text{rep},m}$ from the likelihood $p(y \mid \theta^{(m)})$,
  2. record $T(y, \theta^{(m)})$ and $T(y^{\text{rep},m}, \theta^{(m)})$.

  $\Rightarrow p_B$ is the proportion of indices m for which $T(y, \theta^{(m)}) \leq T(y^{\text{rep},m}, \theta^{(m)})$.
  All this is computationally cheap $\implies p_B$ comes almost for free after posterior sampling is over.
- Caveat: p_B can be used to argue against a model. However, it does not penalize overly complex models, and therefore does not guard against overfitting. Therefore, p_B should not be used as an automatic model selection/rejection tool; human attention is needed.

Predictive comparison - CZK



Summary: Model selection

Bayes factors and marginal likelihoods are

- used for model selection,
- computationally intensive to be computed,
- potentially sensitive to the choice of prior distribution,
- penalize complex models in a natural way (automatic *Occam's razor*).

In this framework,

- models that are compared do not have to have the same set of parameters or be nested,
- decisions can be made involving more than two models,
- external information can be included through the priors.

Variable selection for regression models

Variable selection for regression models

Classical regression model

$$y_i = \mathbf{x}_i\boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \dots, N$$

$\mathbf{x}_i$ ... design matrix / covariates $(1 \times d)$, $\boldsymbol{\beta}$ ... regression parameter

- Which covariates should be included in the model?
- Challenging due to the potentially high number of models to be compared: 2^{d-1} models, where $d = \dim(\boldsymbol{\beta})$ (no model choice for the intercept).

Model selection

Two popular methods for selecting one out of K models $\{\mathcal{M}_1, \ldots, \mathcal{M}_K\}$:

- Penalize models with (too) many parameters (perfect model fit, if $d_k = N$):

$$\text{BIC}_k = \text{model fit} + \text{penalty for number of parameters}$$

- Compute the posterior probability of all models given the data, i.e. $\Pr(\mathcal{M}_1|\mathbf{y}), \ldots, \Pr(\mathcal{M}_K|\mathbf{y})$ using Bayes' rule:

$$\Pr(\mathcal{M}_k|\mathbf{y}) \propto p(\mathbf{y}|\mathcal{M}_k)\Pr(\mathcal{M}_k)$$

  Choose model with highest posterior probability.

- For a regression model, the marginal likelihood and BIC are closely related, if the number of observations is large:

$$\log p(\mathbf{y}|\mathcal{M}_k) \approx \text{-0.5 BIC}_k.$$

Bayesian variable selection

- Two Bayesian computational approaches leading to identical results:
  - **Full model search**: Compute the marginal likelihood and the posterior probability for each model - feasable for 'small' d in normal regression models (e.g., with $d = 200$ regressors $\Rightarrow 1.6 \times 10^{60}$ models $\Rightarrow$ computing time of 5×10^{46} for 1 Mio models per second).
  - **Model space Markov Chain Monte Carlo**:
    - Use a Markov chain to search the model space
    - Sample the various models according to their posterior probability

$\Rightarrow$ Model space MCMC automatically visits the more promising models, models with a very low model probability are not visited in practice;

$\Rightarrow$ Estimate model probabilities by relative frequencies of each model.

SSVS for the standard regression model

stochastic search variable selection

- Define indicators δ_j for each element β_j of $\boldsymbol{\beta}$, $j = 1, \ldots, d$:

$$\begin{array}{ll} \beta_j = 0, & \text{if } \delta_j = 0, \\ \beta_j \text{ unrestricted}, & \text{if } \delta_j = 1. \end{array}$$

$\Longrightarrow$ $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_d)'$ determines the model, e.g.:

$$y_i = x_{i1}\beta_1 + x_{i4}\beta_4 + \beta_5 + \varepsilon_i$$

$$y_i = \delta_1 x_{i1}\beta_1 + \delta_2 x_{i2}\beta_2 + \delta_3 x_{i3}\beta_3 + \delta_4 x_{i4}\beta_4 + \delta_5\beta_5 + \varepsilon_i,$$

with $\boldsymbol{\delta} = (\delta_1, \delta_2, \delta_3, \delta_4, \delta_5)' = (1, 0, 0, 1, 1)'$.

- Each of the 2^d models is represented by a realization of $\boldsymbol{\delta}$ $\Rightarrow$ Inference on $\boldsymbol{\delta}$!
- Priors:
    - Flat prior on β_j.
    - Hierarchical prior on δ_j: $p(\delta_j = 1|\omega_j)$, $\omega_j \sim \mathcal{B}(a, b)$

MCMC Estimation

Stochastic search in the space containing all possible values of $\boldsymbol{\delta} = (\delta_1, \dots, \delta_d)'$ using MCMC

(a) Model search: Sampling of a new indicator $\boldsymbol{\delta}$:
- iterate over j (typically $j = 1, \dots, d$) and sample a new value for δ_j from $p(\delta_j | \boldsymbol{\delta}_{-j}, \mathbf{y})$
$\Rightarrow$ defines a certain restricted regression model

(b) Parameter estimation for the restricted regression model: Sampling of the non-zero regression parameters $\boldsymbol{\beta}^{\boldsymbol{\delta}}$ and the variance σ^2 conditional on $\boldsymbol{\delta} = (\delta_1, \cdots, \delta_d)$ from the posterior distribution

$$p(\boldsymbol{\beta}^{\boldsymbol{\delta}}, \sigma^2 | \boldsymbol{\delta}, \mathbf{y})$$

$\Rightarrow$ standard closed form Bayesian analysis of a regression model

Due to Markov chain theory, all models are sampled according to their posterior probability distribution $p(\boldsymbol{\delta} | \mathbf{y})$ after a suitable burn-in phase.

Metropolis Hastings algorithm

- Propose a new value δ_j^{new} using a proposal $q(\delta_j^{\text{new}}|\delta_j^{\text{old}})$ and accept δ_j^{new} with probability

$$\min\left(1, \frac{p(\mathbf{y}|\delta_j^{\text{new}}, \boldsymbol{\delta}_{\backslash j})p(\delta_j^{\text{new}}, \boldsymbol{\delta}_{\backslash j})q(\delta_j^{\text{old}}|\delta_j^{\text{new}})}{p(\mathbf{y}|\delta_j^{\text{old}}, \boldsymbol{\delta}_{\backslash j})p(\delta_j^{\text{old}}, \boldsymbol{\delta}_{\backslash j})q(\delta_j^{\text{new}}|\delta_j^{\text{old}})}\right).$$

- The MH algorithm typically proposes to switch from δ_j^{old} to $\delta_j^{\text{new}} = 1 - \delta_j^{\text{old}}$, in which case the proposal ratio cancels.
- The MH accepts a new model, whenever its posterior probability

$$\Pr(\boldsymbol{\delta} = (\delta_j^{\text{new}}, \boldsymbol{\delta}_{\backslash j})|\mathbf{y}) \propto p(\mathbf{y}|\delta_j^{\text{new}}, \boldsymbol{\delta}_{\backslash j})p(\delta_j^{\text{new}}, \boldsymbol{\delta}_{\backslash j})$$

is larger than for the old model.

Posterior analysis

Also called Bayesian Model Averaging (BMA)

1. Posterior probability for the indicator to be non-zero:

$$Pr(\delta_j = 1|\mathbf{y}) = \frac{1}{M}\sum_{m=1}^{M}\delta_j^{(m)}$$

2. Bayesian model selection
   - Median probability model: take effects with $Pr(\delta_j|\mathbf{y}) > 0.5$.
   - Highest probability model: $\boldsymbol{\delta}$ most often visited during MCMC.
3. Bayesian model averaged parameter estimates:

$$\hat{\beta}_j = \frac{1}{M}\sum_{m=1}^{M}\beta_j^{(m)}$$

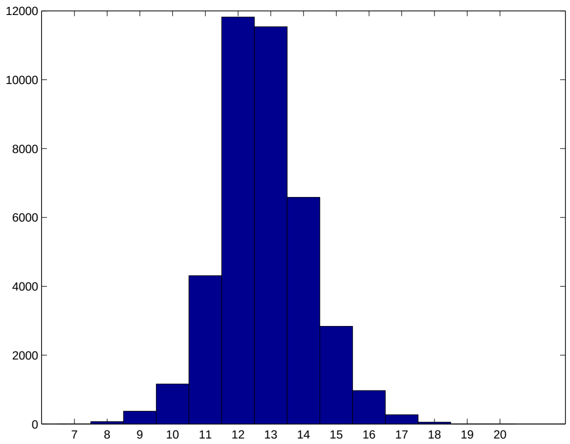
average of betas from different models (with different covariates included) -> nice estimate (robust estimation of the true effect regardless of other covariates)

Product Rating Data

- 213 Austrian consumers rated a product (mineral water) on a 20 point rating scale
- Each consumer evaluated 15 brand (Römerquelle, Vöslauer, Juvina, Waldquelle, Kronsteiner) - price (low, medium, high) combinations
- Additionally we know 7 consumer specific attributes: education, income, gender, age, price sensitivity, brand loyalty, time used for buying mineral water
- If we add all 105 interaction effects between product and consumer specific attributes, this yields 120 variables in the design matrix, i.e. there are $2^{119} = 6.6 \cdot 10^{35}$ possible models.
- We run the MCMC scheme for 50 000 iterations and skip 10 000 iterations for burn-in. 18 757 different models were visited.

Spying on the indicators

Histogram of the number of unconstrained elements in β during MCMC (max: 120):



Posterior analysis

	$\hat{\beta}$	$\Pr(\delta_j = 1 \mid \mathbf{y})$	Median-Pr-Model	Highest-Pr-Model
const	14.48	1	x	x
brand 1 (RQ)	4.14	1	x	x
brand 2 (VOE)	4.46	1	x	x
brand 3 (JU)	.04	.03		
brand 4 (WA)	.01	.03		
price	-1.7	1	x	x
price, quadratic	-.0	0.01		
education · JU	-1.45	0.79	x	x
age · JU	0.03	0.69	x	x
age · WA	0.02	0.65	x	x
⋮	⋮	⋮	⋮	

Shrinkage priors in regression analysis

Bayesian variable selection

- Last section:
    - Full model search computing the marginal likelihoods
    - Stochastic search: model space MCMC
- Third alternative: Shrink small regression coefficients to zero: shrinkage priors
    - Peak/singularity at zero, heavy tails.

    - e.g. Normal gamma prior (Griffin and Brown, 2010), Horse-shoe prior (Carvalho et al, 2010)

Effect of the prior distribution i

- What is the effect of a prior on the estimation of θ?

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)p(\theta)$$

- Normal regression model:

$$\begin{aligned} y_i &= \mathbf{x}_i\boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}\left(0, \sigma^2\right). \\ \boldsymbol{\beta}|\sigma^2 &\sim \mathcal{N}\left(0, \phi^2\mathbf{I}\sigma^2\right) \end{aligned}$$

- The normal prior on the regression effects induces shrinkage of $\boldsymbol{\beta}$ to zero:

$$\boldsymbol{\beta}|\mathbf{y}, \sigma^2 \sim \mathcal{N}\left((\mathbf{X}'\mathbf{X} + \frac{1}{\phi^2}\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}, \sigma^2(\mathbf{X}'\mathbf{X} + \frac{1}{\phi^2}\mathbf{I})^{-1}\right)$$

Ridge regression

- Posterior mean of β corresponds to Ridge estimate:

$$(\mathbf{X}'\mathbf{X} + \frac{1}{\phi^2}\mathbf{I})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X} + \boldsymbol{\lambda}\mathbf{I})^{-1}\mathbf{X}'\mathbf{y} = \hat{\beta}^{Ridge}$$

  where the prior variance is the penalisation parameter which controls shrinkage.

- Ridge regression minimizes the penalized sum of squared residuals:

$$||\mathbf{y} - \mathbf{X}\beta||^2 + \lambda||\beta||^2$$

$\Rightarrow$ Prior on β is a penalty - general pattern!

Regularization in regression analysis

- Regression model: $y_i = \mathbf{x}_i\boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}\left(0, \sigma^2\right).$
- LASSO [Tibshirani, 1996] - penalized sum of squared errors:

$$\sum_{i=1}^{N}(y_i - \mathbf{x}_i\boldsymbol{\beta})^2 + \lambda^2 \sum_{j=1}^{d} |\beta_j|$$

- Bayesian LASSO [Park and Casella, 2008] - log posterior $\log p(\boldsymbol{\beta}|\mathbf{y}, \sigma^2, \lambda^2)$ under prior $\beta_j \sim \mathrm{Lap}\left(0, 2\sigma^2/\lambda^2\right)$:

$$-\frac{0.5}{\sigma^2}\left(\sum_{i=1}^{N}(y_i - \mathbf{x}_i\boldsymbol{\beta})^2 + \lambda^2 \sum_{j=1}^{d} |\beta_j|\right) + \mathrm{constant}$$

The Laplace prior has a sharp peak at zero, which encourages sparse solutions (many coefficients set exactly to zero). => either the beta is 0 (very small) or large, nothing between

Priors as Penalties

- **Penalty Parameter (λ)**: In frequentist regularization, λ controls shrinkage. In Bayesian terms, λ is directly related to the inverse variance of the prior. A high penalty (λ) corresponds to a prior with small variance, reflecting a strong belief that the coefficients are zero.
- **Uncertainty Quantification**: Unlike frequentist regularization, which provides only a point estimate (the best β), Bayesian regularization provides a full posterior distribution, allowing for the calculation of uncertainty around parameter estimates.
- **Automatic Tuning**: Bayesian methods allow the use of hierarchical priors, where hyperparameters (like the penalty strength) are estimated from the data, acting as 'automatic cross-validation'.
- **Alternative Priors**: Beyond Gaussian and Laplace, Bayesian regression can use other "shrinkage" priors such as the Horseshoe or Spike-and-Slab priors, which can better manage the tradeoff between shrinking small coefficients and leaving large ones unaffected.

Shrinkage priors in Bayesian regression analysis

Hierarchical representation as scale mixtures:

- **Bayesian Lasso prior** [Park and Casella, 2008]:

$$\beta_j|\tau_j^2 \sim \mathcal{N}\left(0, 2/\lambda^2\tau_j^2\right), \qquad \tau_j^2 \sim \mathcal{E}\left(1\right).$$

- **Normal-Gamma prior** [Griffin and Brown, 2010]:

$$\beta_j|\tau_j^2 \sim \mathcal{N}\left(0, 2/\lambda^2\tau_j^2\right), \qquad \tau_j^2 \sim \mathcal{G}\left(a^\tau, a^\tau\right), \quad a^\tau < 1.$$

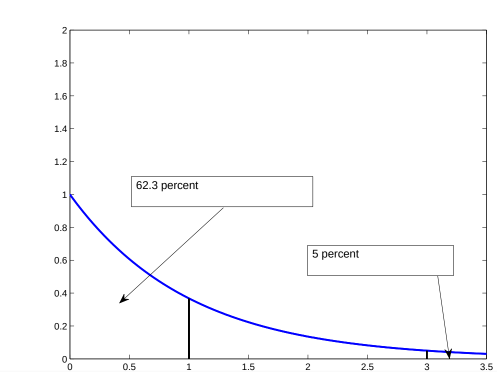
  $a^\tau = 1$ leads to the Bayesian Lasso model, $a^\tau < 1$ introduces more sparsity.

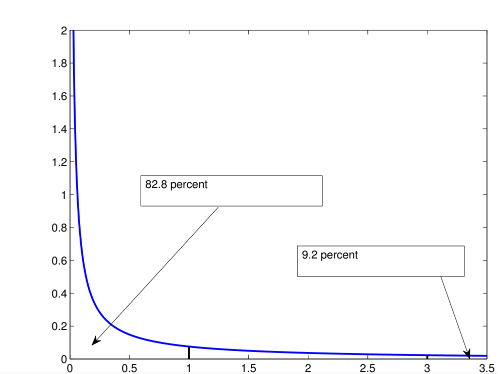
- Special cases of **global-local shrinkage priors** [Polson and Scott, 2011]:
  - shrink globally: $\mathrm{Var}(\beta_j) = \mathrm{E}(\beta_j^2) = 2/\lambda^2$;
  - act locally for each coefficient: $\tau_j^2 < 1$ – more shrinkage, $\tau_j^2 > 1$ – less shrinkage.
- **Learn τ_j^2 from the data!** apriori: E[tau_j]=1 -> look at the data to get aposteriori E[tau_j]

  $\rightarrow$ Hierarchical representation allows for efficient Gibbs sampling.

Shrinkage priors

Shrinkage prior for τ_j^2 based on the exponential distribution (left) and the Gamma prior with $a^\tau = 0.1$ (right)



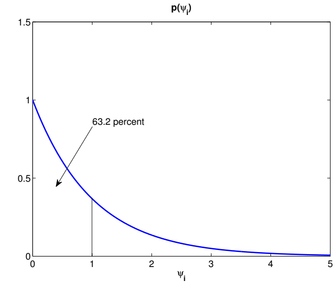


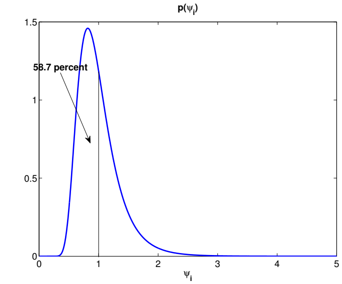
Shrinkage versus non-shrinkage prior

- Not every scale-mixture prior introduces sparsity, e.g. Student-t prior $\beta_j|\nu, \sim t_{2\nu}(0, \lambda^2)$:

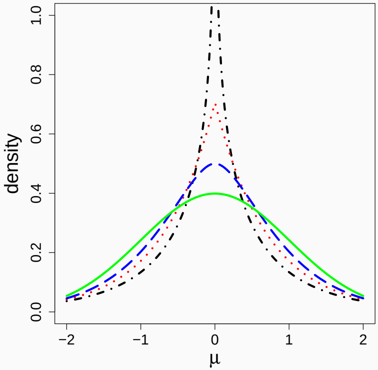
$$\beta_j|\tau_j^2 \sim \mathcal{N}\left(0, \lambda^2\tau_j^2\right), \qquad \tau_j^2 \sim \mathcal{G}^{-1}(\nu, \nu).$$

- While this prior has heavy tails, it does not encourage shrinkage toward 0, see right-hand side of following figure ($\nu = 5$).





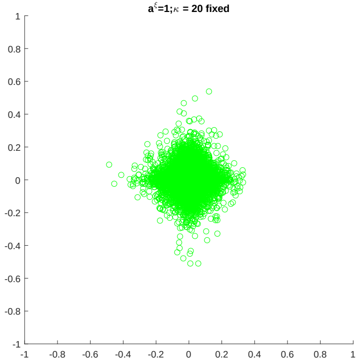
Comparing priors for β_j (marginalized over τ_j^2)

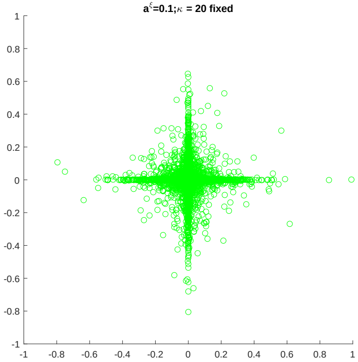


Standard normal (green solid line), Student-t with $\nu = 5$ (blue dashed line), Bayesian LASSO prior (red dotted line), normal gamma prior for $a^{\tau} = 0.5$ (black dot-dashed line)

Comparing bivariate priors (marginalized over τ_j^2 and τ_ℓ^2)

Simulations of (β_j, β_ℓ) for different coefficients from a bivariate Lasso prior (left) and a bivariate normal Gamma prior with $a^\tau = 0.1$ (right); $\lambda^2 = 20$ for both priors.





MCMC Estimation under normal-Gamma prior

(a) Sample $\boldsymbol{\beta}$ conditional on σ^2 and the prior variances $\boldsymbol{\tau} = (\tau_1^2, \dots, \tau_d^2)$:
- use **conditionally Gaussian prior** $\mathbf{B}_0 = \mathrm{Diag}\left(\tau_1^2 \dots \tau_d^2\right)$;
- $\boldsymbol{\beta}|\boldsymbol{\tau}, \sigma^2, \mathbf{y} \sim \mathcal{N}_d\left(\mathbf{b}_N, \mathbf{B}_N\right)$ is multivariate normal distribution with moments given by (week 2, slide 9).

(b) Sample from σ^2 from $p(\sigma^2|\boldsymbol{\beta}, \mathbf{y})$ which is an inverted Gamma distribution $\mathcal{G}^{-1}\left(c_N, C_N\right)$.

(c) NEW: Sample the prior variances $\tau_j|\beta_j, \lambda^2$ for $j = 1, \dots, d$ from conditionally independent **generalized inverse Gaussian distributions**

The generalized inverse Gaussian (GIG) distribution

- Derive the posterior $p(\tau_j^2|\beta_j) \propto p(\beta_j|\tau_j^2)p(\tau_j^2)$:
  - the conditionally normal prior $\beta_j|\tau_j^2 \sim \mathcal{N}\left(0, \tau_j^2\right)$ leads to a likelihood which is the **kernel of an inverted Gamma density** in τ_j^2:

$$p(\beta_j|\tau_j^2) \propto (\tau_j^2)^{-1/2} \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right).$$

  - the **Gamma prior** $\tau_j^2 \sim \mathcal{G}\left(a^\tau, a^\tau\lambda^2/2\right)$ is not conditionally conjugate:

$$p(\tau_j^2) \propto (\tau_j^2)^{a^\tau-1} \exp\left(-\frac{a^\tau\lambda^2\tau_j^2}{2}\right).$$

- Combining the likelihood $p(\beta_j|\tau_j^2)$ with the prior $p(\tau_j^2)$ yields:

$$p(\tau_j^2|\beta_j) \quad \propto \quad (\tau_j^2)^{(a^\tau-1/2)-1} \exp\left(-\frac{a^\tau\lambda^2\tau_j^2}{2}\right) \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right).$$

The generalized inverse Gaussian (GIG) distribution

- The inverse Gaussian distribution, $\mathcal{GIG}\left(p, a, b\right)$ is a three-parameter family with density ($a > 0$, $b > 0$ and p is a real parameter):

$$f(y) = \frac{(a/b)^{p/2}}{2K_p(\sqrt{ab})} y^{p-1} e^{-(a/2)y} e^{-b/(2y)},$$

where $K_p(z)$ is the modified Bessel function of the second kind.

- The conditional posterior $p(\tau_j^2|\beta_j)$ is a **generalized inverse Gaussian (GIG) distribution** with $p = a^\tau - 1/2$, $a = a^\tau \lambda^2$, $b = \beta_j^2$:

$$\tau_j^2|\beta_j \sim \mathcal{GIG}\left(a^\tau - 1/2, a^\tau \lambda^2, \beta_j^2\right).$$

- A **very stable generator** is implemented in the R-package `GIGrvg` by J. Leydold [Hörmann and Leydold, 2014]

Adding another hierarchy: Hierarchical normal gamma prior

- Normal gamma prior:

$$\beta_j | \tau_j^2, \lambda^2 \sim \mathcal{N}\left(0, 2/\lambda^2 \tau_j^2\right), \qquad \tau_j^2 | a^\tau \sim \mathcal{G}\left(a^\tau, a^\tau\right), \quad a^\tau < 1.$$

- Add a **hyper prior for global shrinkage parameter λ^2**:

$$\lambda^2 \sim \mathcal{G}\left(e_1, e_2\right).$$

  - Introduces dependence between the coefficients $\beta_1, \ldots, \beta_d$ (we don't believe that all coefficients are non-zero).
  - **Add sampling step to MCMC scheme** to update $\lambda^2 | \beta_1, \ldots, \beta_d, \tau_1^2, \ldots, \tau_d^2$ (Gamma distribution).
  - One can learn $a^\tau | \boldsymbol{\tau}$ given $\boldsymbol{\tau} = (\tau_1^2, \ldots, \tau_d^2)$ as well (see later).

Application: Inflation modelling

Generalized Phillips curve [Belmonte et al., 2014]:

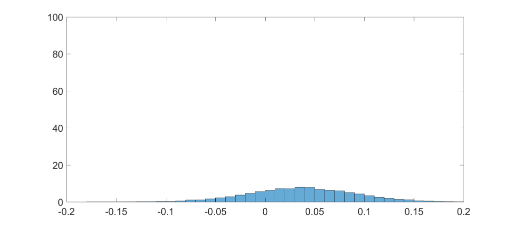
$$y_{t+h} = c + \sum_{j=0}^{p-1} \phi_j \cdot y_{t-j} + \boldsymbol{\alpha} \mathbf{x}_t + s_t + \varepsilon_{t+h},$$

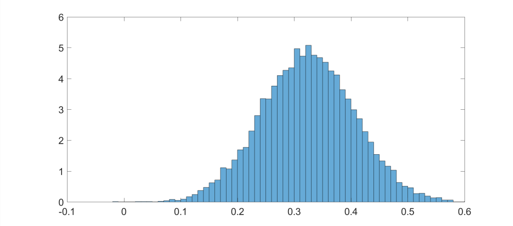
Monthly data from February 1994 to November 2010 ($T = 190$), 37 coefficients:

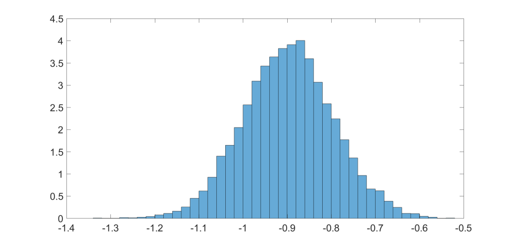
- lags of inflation ($p = 12$) and 11 monthly seasonal dummies
- 13 other predictors $\mathbf{x}_t$ (unemployment rate, 1-month and 1-year Euribor, change in industrial production index, change in loans, change in monetary aggregate M3, oil price, ...)

Standard Bayesian inference

Posterior of β_j for oil (24), changes in book levels (25), and unemployment (26)

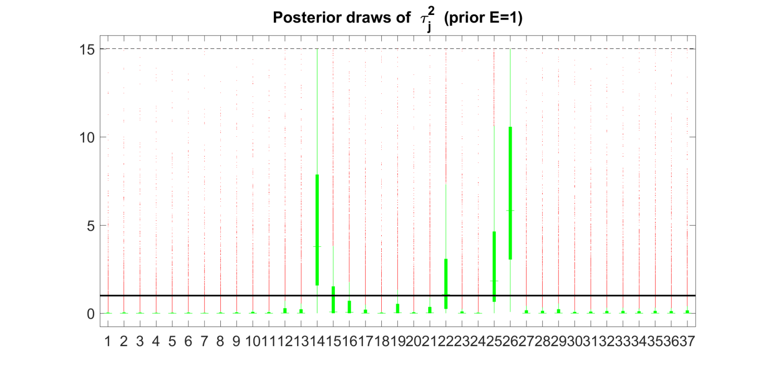






Bayesian inference with shrinkage priors

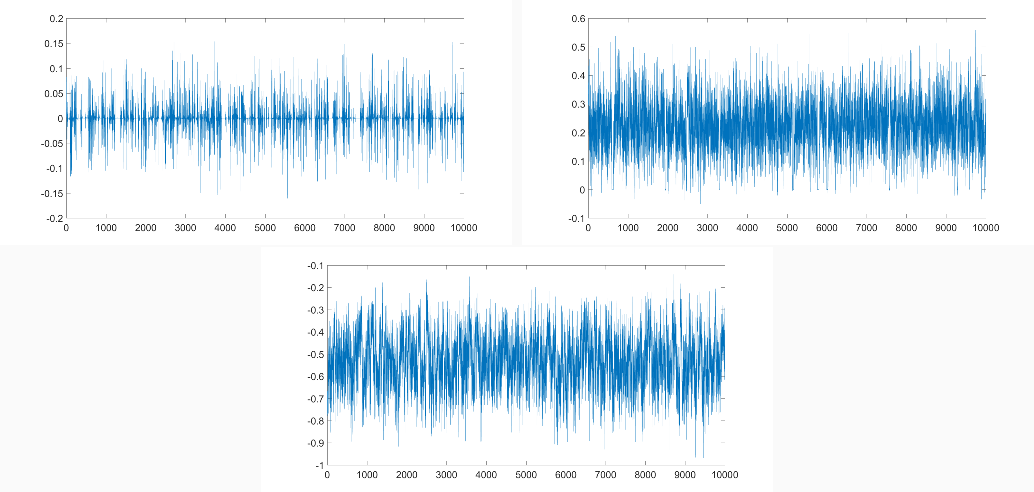
Posterior distribution of τ_j^2 (hierarchical shrinkage prior with $a^\tau = 0.1$, $e_1 = e_2 = 0.001$)



1-M (14), 1-Y Euribor (15), unemployment (26); % change M3 (22), order book level (25)

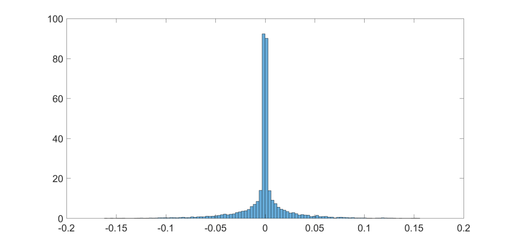
Bayesian inference with shrinkage priors

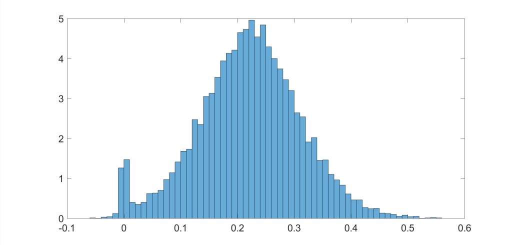
MCMC draws of β_j for oil (24), changes in book levels (25), and unemployment (26)

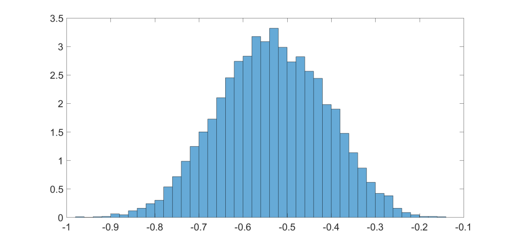


Bayesian inference with shrinkage priors

Posterior of β_j for oil (24), changes in book levels (25), and unemployment (26)







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